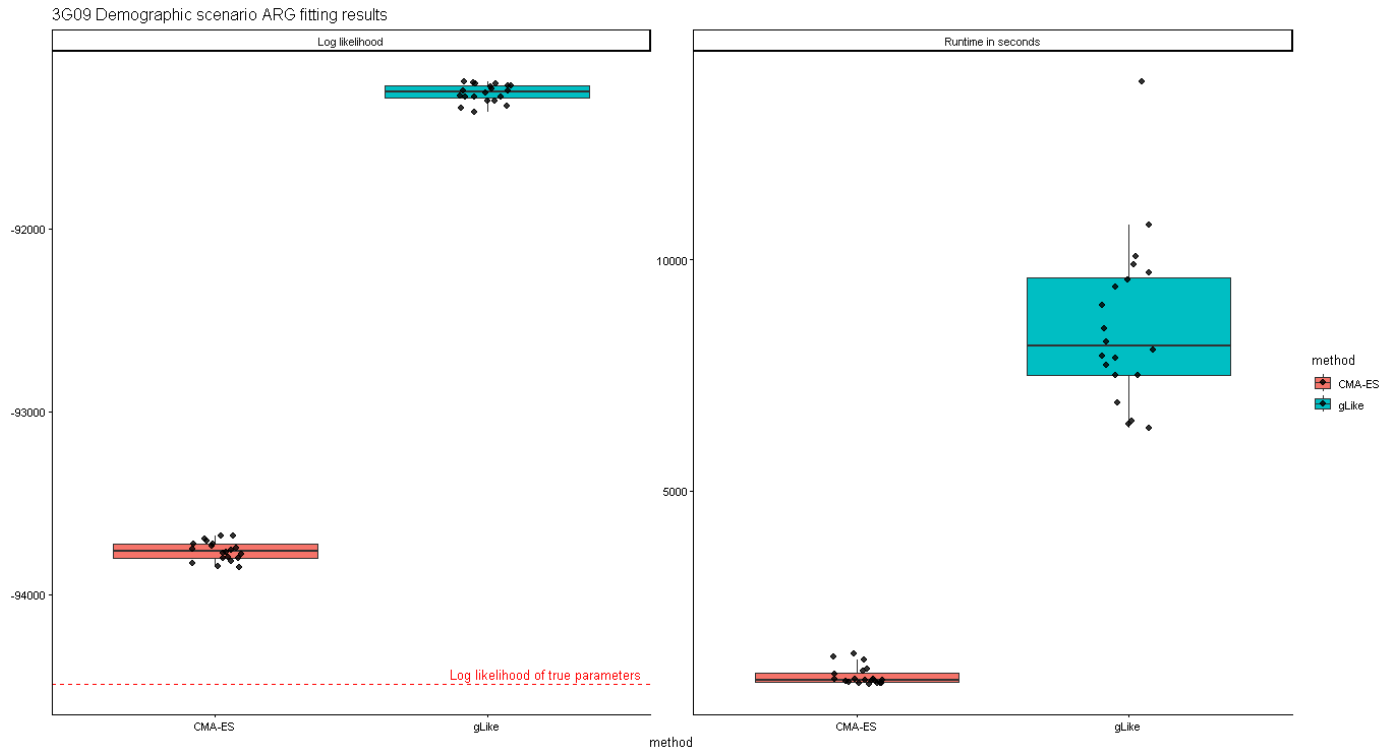


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Summary

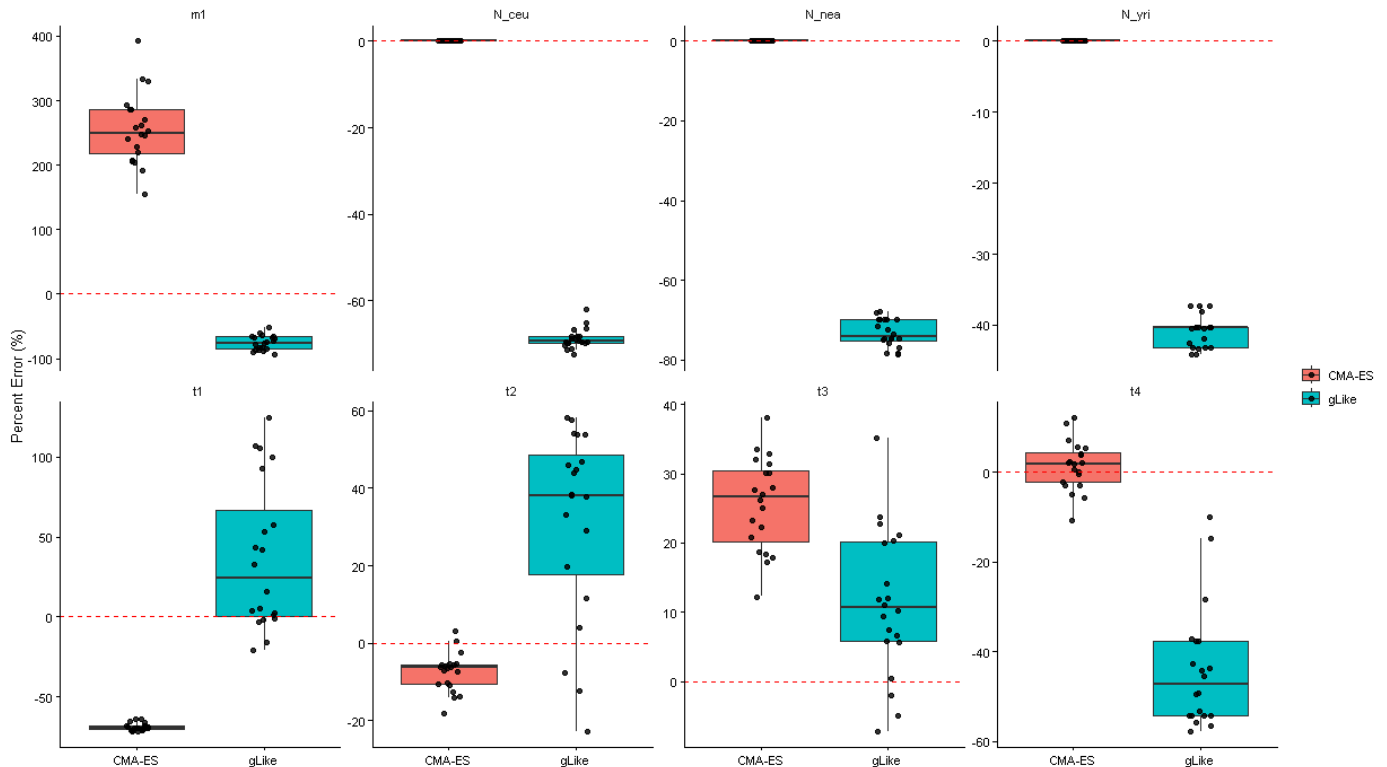
- I proceeded with 10 trees
- I fixed the implementation of CMA-ES
 - NOTE: CMA-ES as currently implemented requires a pre-specified set of boundary conditions for models in which such conditions exist (e.g., migration rate, admixture proportions, etc.), otherwise the program will explore non-feasible states
 - NOTE: A present drawback of my implementation of CMA-ES is that the "tolerance" for a given iteration is the same for all parameters, i.e., at some point the iterator no longer wants to move because it believes things to be at equivalent log likelihood. However, the same tolerance point is used for all parameters, even when parameters have drastically different scaling, e.g., migration rate vs. $N_{\text{Ancestral}}$
 - I chose a tolerance of 0.05 arbitrarily
- I compare benchmarking for $\log p(x)$ and runtime between the original gLike method and CMA-ES
- I compare parameter fit relative to the true underlying parameters for the original gLike method and CMA-ES
- In all cases, I provided the same underlying simulation (as determined by seed) and same underlying true tree
- In all cases, I used 10 trees, 20 replicates, and the same initial starting conditions

gLike vs. CMA-ES (computational benchmarking)



- - CMA-ES data shown in red
 - CMA-ES seems to consistently outperform in terms of both having a shorter runtime and a better log likelihood
 - CMA-ES took on average around 15 minutes to run
 - gLike original method took on average around 2 hours to run

gLike vs. CMA-ES (parameter)



- CMA-ES data shown in red
 - CMA-ES overestimates $m1$, the migration rate, and $t3$, while underestimating $t1$. It is otherwise fairly accurate at recovering the true simulation parameters
 - gLike underestimates $m1$, all of the effective population sizes, and $t4$, while overestimating $t2$ and $t3$. It has a broad range of bidirectional percentage error across different parameters

TODO

- Consider NH demographic scenario (I actually almost have this data)

```
(glike) bash-4.4$ squeue -u mahj
JOBID PARTITION NAME USER ST TIME NODES NODELIST(REASON)
11642140_18 main benchmar mahj R 6:01:50 1 d11-22
11642140_3 main benchmar mahj R 6:04:04 1 d05-26
11642140_4 main benchmar mahj R 6:04:04 1 d11-23
11642140_8 main benchmar mahj R 6:04:04 1 d11-19
11659223 main interact mahj R 10:29 1 a02-02
11658625 main interact mahj R 21:34 1 d05-40
```

- : Four replicates left for NH demographic scenario fit by gLike .
 - CMA-ES on average took about 17 minutes to run
 - For jobs that have finished, gLike original method took about 5 hours to run
- Look into parameter scaling transform so that parameter magnitude doesn't interfere with
- Take three populations: 3G09, NH (or American Admixture), ancient European (see Caoqi's paper)

- Remove migration parameters / set to 0 --> will this fix the percentage error for estimation of t_1 , t_2 , t_3 , t_4 ?
- Diagnostic plots (talk to Ji Tang) which show parameter trajectory over time